

Solving simultaneous equations

Key Points

To solve simultaneous (linear) equations we can write them as a matrix equation $Ax = b$

Then if A is invertible we get a unique solutions

Otherwise we get infinitely many if the equations are consistent and no solutions if they are inconsistent

To actually find the unique solutions we can pre multiply by the inverse.

Linear equations in x, y, z represent planes

Basic Skills

Q1. For the matrix $M = \begin{pmatrix} 3 & 4 \\ 5 & -2 \end{pmatrix}$

a) Find the determinant $|M|$

b) Find the inverse M^{-1}

c) Solve the equations $3x + 4y = 5, 5x - 2y = 20$ by solving the matrix equation $Mx = b$ picking an appropriate vector b

Q2. Using an appropriate matrix explain why there are no solutions to the equations $2x + y = 2, 6x + 3y = 7$

Q3. Using an appropriate matrix explain why there are infinite solutions the equations $x + 4y = 2, 2x + 8y = 4$ and state all the solutions

Competency Skills

Q4. For the three simultaneous equations:

$$\begin{aligned} 5x + 2y - z &= 8 \\ 3x - y + z &= 15 \\ x + 3z &= 18 \end{aligned}$$

a) Write the matrix form equation

b) Explain how we know that these three simultaneous equations have a unique solution

c) Show that the matrix inverse is $A^{-1} = \frac{1}{32} \begin{pmatrix} 3 & 6 & -1 \\ 8 & -16 & 8 \\ -1 & -2 & 11 \end{pmatrix}$

d) Hence, or otherwise, find the unique solution

Q5. Solve the matrix equation (or state it is not possible to solve) for the 3 following cases:

$$\begin{pmatrix} 6 & -1 & 2 \\ a & 2 & -4 \\ 6 & 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b \\ -6 \\ 0 \end{pmatrix} \quad \text{a) } a = 5, b = 20 \quad \text{b) } a = -12, b = 3 \quad \text{c) } a = -12, b = 6$$

Advanced Skills

Q6. Draw a flowchart which can be used to determine if a matrix equations has unique, infinite or no solutions

Q7. For the three planes $\pi_1: 5x + 3y - 2z = 1, \pi_2: 6x + 2y + 3z = 5$ and $\pi_3: 7x + y + 8z = 5$

a) Determine if the planes all intersect at any unique point

b) By eliminating y show the three planes form a prism of planes

Q8. Determine if for the following set of simultaneous equations:

There is some k such that there is a unique solution and if so state all k

$$\begin{aligned} 3x + 2z &= 6 \\ x + ky + 5z &= 1 \\ -x + 2y + kz &= -6 \end{aligned}$$

Extension

Q9. In the case of three planes describe all possible scenarios of their joint intersection

Hint: For the infinite solutions case consider multiple options to 'how infinite' the solutions are